Examiner reports

Q1.

Students did well here, with over 80% of students identifying the correct option. There was no clear second choice.

Q2.

For many students part (a) was a straightforward application of a practised routine:

- $F = ma$ applied separately to each particle
- T eliminated from the two equations
- Use of $s = ut + ½at^2$ to complete the solution.

A minority started with a single equation, $1.8g - 1.2g = ma$, which received zero marks. An equation separated from its underlying mechanical principle is hard to grasp, as evidenced by more of these students using $m = 1.8$ rather than $m = 3$.

Some students chose the wrong *suvat* equation and had to apply a second to find t. Premature or even final rounding sometimes lost the accuracy mark, where 3 significant figures were expected.

Students frequently approached the demand in part (b) in the wrong way. They should understand that what is required is an assumption that is necessary, sensible, specific and not already stated in the question. Thus saying that there was “no friction at the peg” or that “the string had negligible mass” scored zero marks. Similarly “no other forces act” is too general. A minority realised the necessity of sufficient length of string.

It is often better if the answer starts with an “I have assumed that” to ensure that the assumption goes on to be properly explained. For example “I have assumed that there is no air resistance” is better than just “air resistance” where the assumption is not clear.

Q3.

In part (a), most students recognised the effect of the air resistance, although some thought that Jason was behind because he was not applying enough driving force. Since “Less air resistance for Jason” and “More air resistance for Laura” were both correct answers, an unattributed “Less air resistance” did not explain sufficiently.

For part (b), most students correctly calculated the deceleration for Laura, but some had this as a positive acceleration. Some had an unexplained F in their equation, $F - 40 = 64a$, and made little progress. Others replaced this F with 64 or $64g$, getting a positive a.

It was possible to complete the question in a variety of ways:

- use $v^2 = u^2 + 2as$ and show $s < 40$ when $v = 0$, $a = -0.625$ and $u = 6.944$
- use $v^2 = u^2 + 2as$ with $a = -0.625$ and show that $v = 0$ at $s = 40$ as long as $u < 7.07$
- use $v^2 = u^2 + 2as$ with $v = 0$, $s = 40$ and $u = 6.944$ and deduce that $a < -0.603$
- as above but go back one stage to show resistive force must be > 38.5 N
- use $v^2 = u^2 + 2as$ with $a = -0.625$, $s = 40$ and $u = 6.944$ to get $v^2 = -1.8$ and deduce that Laura had come to rest before this.

The first method was the most popular, but clear solutions using all other methods were seen. All of these methods lost accuracy marks when 25 was used as the value of u.

For part (c), the assumptions needed to be necessary, sensible, specific and not already stated in the question. Since the total resistive force was given, this was one case where “air resistance” was not acceptable. A minority recognised that reaction time had been ignored, or that the resistive force might diminish as Laura slowed. Many stated that they had assumed there was no friction (between bike and road), when in fact friction here is essential for the brakes to work.

Q4.

The majority recalled that integration was required in part (a), although some differentiated and a few used $s = vt$. A large proportion either forgot the constant of integration or did not calculate its value, losing half the marks.

In part (b), the time of 2 seconds ensured that the toy was stationary when it launched the ball. Many, however, tried to bring extraneous distances and velocities into this question. Making the model fit the situation is an important skill to be assessed. For those who did this, the actual calculation was usually carried out correctly.

Q5.

This question was well answered with approximately two thirds of the students selecting the correct response. The most frequently chosen incorrect response was 0.71 m s^{-2}.

Q6.

A high proportion of students had great difficulty in forming an appropriate equation of motion for part (a)(i). Those who attempted to treat the system as a whole made mistakes such as using the wrong total mass, including incorrect tension forces, using weight instead of mass or forgetting to include the resistance on the buggy. Some students considered the skater and buggy separately and similar errors were seen, however a fairly common error was to subtract the mass of the skater from the mass of the buggy in the equation of motion of the buggy.

Those students who wrote down a correct equation of motion usually completed the question successfully. Approximately half of the students scored full marks on part (a)(i).

In part (a)(ii) students were still able to make progress even if they had gone wrong in (a)(i). Provided their equation of motion was correct using their R they were given credit. Over half of students scored at least 2 marks.

Part (b) was a request for assumptions in a very standard model. A significant number of students gave assumptions that did not need to be made, as they were given in the question. For example, it was unnecessary to assume the road was flat as the question stated “horizontal road”. Students should also avoid words like “flat” or “straight” when they mean “horizontal”. A correct assumption would be that the rope was horizontal.

In part (c)(i), students who had gone wrong in earlier parts of the question were still able to make good progress and they were credited for using correct techniques. Correctly using their incorrect R from part (a)(ii) often resulted in 4/5 marks. Common errors were caused by students over complicating the problem, often attempting, unnecessarily, to find the time taken for either the buggy or skater to stop.

Over half of students made some progress on (c)(ii), but the general standard of explanation was poor. A significant number of students showed a misunderstanding of the forces model, or may have been trying to over-explain. Statements like "the rope has been dropped so the buggy no longer feels the resistance from the skater" are not accurate. It is more concise and accurate to simply say "there is no tension acting on the buggy".

Q11.

This question was done very well by the vast majority of students with many gaining full marks. Minor mistakes were the main reasons that students lost marks. Occasionally, students treated the car as if it was moving with constant velocity, for example using Time $= \frac{\text{Distance}}{\text{Speed}}$ or similar. There were a few cases where students quoted constant acceleration equations incorrectly. Also there were a few, but not many instances, in part (c) where students found the mean of two speeds.

Q12.

Parts (a) and (b) of this question were frequently done well. In part (b) some students omitted the initial position. An unusual error that was seen a number of times was to write something like $13.6\mathbf{i} = (6\mathbf{i} + 2.4\mathbf{j})t + \frac{1}{2}(-0.8\mathbf{i} + 0.1\mathbf{j})t^2$ often followed by $(6\mathbf{i} + 2.4\mathbf{j})t + \frac{1}{2}(-0.8\mathbf{i} + 0.1\mathbf{j})t^2 - 13.6\mathbf{i}$

Part (c) was generally not done well. A great many students tried to use the position vector instead of the velocity vector. Of those who did realise that the velocity vector was required a few did not introduce a negative sign and simply equated the components of the velocity as if the direction of travel was north-east rather than north-west. Also a few students who made good progress stopped when they had obtained $28\mathbf{i} + 36\mathbf{j}$ and did not find the magnitude of this to obtain the distance as requested in the question.

Q13.

This question was also done very well with a high success rate. The main issues that prevented students gaining full marks were arithmetical errors or reading incorrect values from the graph. A few students omitted one section when finding the area. Occasionally, students would incorrectly take the mean of some velocities, for example calculations like $\frac{5+6+7}{3} = 6$, but this could not be described as a common approach.

Q14.

Part (a) was a very standard question and many students did well with this part of the question. However, a small number of students ignored the instruction to form two equations of motion. In part (b), some students did not convert the 40 cm to metres and obtained speeds that related to moving 40 m rather than 40 cm. These students did gain some marks, but lost the final accuracy mark.

In part (c) a common error was to use the wrong acceleration, with some students continuing to use 4.9 m s^{-2} rather than −9.8 m s^{-2}. In this part there was also some confusion between the values to use for u and v, with some students taking u as zero at the maximum height.

Most students were able to say that the acceleration would be reduced, when answering part (b), but a much smaller proportion of the students could give a good explanation.

Those who mentioned the effect of the air resistance on the resultant force normally gained both marks.

Q15.

Part (a) was the most accessible part of this question. While there were a good number of correct responses, there were also a significant number of students who had problems with the initial position. Often this was initially placed on the wrong side of the equals sign so that the students ended up with $-(500\mathbf{i} + 200)$ in their expressions for the position vector.

Part (b) was much more challenging for the students. Those who were able to form two quadratic equations from the components of the position vector were usually able to go on and solve these to find the correct time. Incorrect approaches included trying to work with the magnitude of the position vector or not using the position vector of the rock.

Part (c) was also problematic for some students. Many ended up with $\dfrac{300\mathbf{i}+600\mathbf{j}}{\text{time}}$ instead of $\dfrac{-300\mathbf{i}-600\mathbf{j}}{\text{time}}$.

The explanations in part (d) were generally inadequate, with answers like velocity has direction and speed does not, which did not go far enough to justify the award of the mark.

Q16.

(a) Almost all students were familiar with the equations of motion of a projectile and they were able to use them to find the equation of the projectile's trajectory in terms of x, u, g and $\tan\vartheta$.

(b) (i) Most students interpreted the context of the question correctly in relation to the given co-ordinates axes and they were able to make the appropriate substitutions into their equation of the trajectory. They were then able to solve the quadratic equation in $\tan\vartheta$ However, many students lost the last accuracy mark because they stated the given values of ϑ without writing down the unrounded values to at least four significant figures.

(ii) This part proved more challenging for some students. Those who answered this part correctly were able to find the horizontal and vertical components of the velocity of the ball at the instant of reaching the basket and then used trigonometry to find the required angle. Students needed to specify the direction of the motion of the ball by giving a correct acute angle and stating whether the angle was made with the horizontal or the vertical.

Q17.

There were very many good solutions to this question, with many candidates gaining full marks. There were some arithmetic errors that were introduced in parts (a) and (b) when calculating the distances. The questions about the displacement and the average velocity in parts (d) and (e) did cause some confusion, particularly for some candidates who tried to use the formula $s = \dfrac{1}{2}(u + v)t$.

Q18.

There were very many good attempts at both parts of part (a) with a lot of correct answers. However, there was quite a bit of confusion with part (b). A very common wrong answer was 686 N, and it appears that there was some confusion between the words 'resultant' and 'reaction'. There were some good solutions to part (c), although this part of the question was more demanding. A considerable range of incorrect responses were seen, including the omission of g, as in $70 \sin \alpha = 58.8$, the use of cos instead of sin or the use of the normal reaction from part (b), as in $686 = 70g \sin \alpha$.

In the second part of part (c), it was not unusual to see the 30 N force with the wrong sign which produced the following incorrect equation: $70\, g \sin \alpha = 58.8 - 30$. In part (d), there were many good answers where the candidates linked the resistance to the speed of the cyclist. However, some simply said things such as "the resistance will not be constant" and did not give a reason to support this answer.

Q19.

There were many good attempts at this question. In part (a)(i), many candidates were able to obtain the correct answer although some found the time and then the acceleration. There were only a few cases of candidates confusing u and v in their equations. In part (a)(ii), there were again many good responses, but in some cases inappropriate information was used, for example using a distance of 75m or a final speed of 10m s–1. In part (a)(iii) there were again many correct answers, but quite a few candidates did not find the magnitude of the force and gave negative answers. This was also the case in part (b), where candidates also sometimes added the 200 to their previous answer instead of subtracting it from it.

Q20.

There were some vary varied responses to this question. Part (a) was often done well and the candidates who attempted to find the position vector generally gained the marks available. However, quite a few gave the velocity instead of the position vector. Those who had a velocity vector in part (a) then generally used this in part (b) and scored no marks. Some who had a correct position vector in part (a), used the wrong component to try to answer part (b).
Part (c) was altogether more challenging and many candidates worked with position vectors rather than velocity vectors. This clearly was the wrong approach, but some expended a lot of effort solving the quadratic equations that they produced. Some candidates came up with the correct velocity, but without showing how they had obtained their answer. It is important that working is justified. Sometimes a trial and improvement method led candidates to the correct answer.

Q21.

There were many correct answers to part (a) and the candidates were clearly helped by the presence of the printed answer. In most cases the arguments given were sound, but in others this was not the case. For example some candidates simply wrote statements such as $22.4 \sin \theta - 19.6$ and never included an equals sign or a zero. This was regarded as an important omission.
There were also lots of good answers to part (b), but some basic errors, such as the omission of a negative sign with g or the use of sin(0875). In part (c) there were also some good answers. Some candidates did not realise that the time of flight could be found by doubling the 2 seconds that was given, and did quite a lot of work finding the time of flight.

In some cases there were errors in the calculation of the distance. For example an incorrect application of $s = \frac{1}{2}(u + v)\,t$ was seen with statements such as $s = \frac{1}{2}(10.8 + 0) \times 4$.
The approaches to part (d) were many and varied. There were quite a number of good responses, but often candidates did not really get very far with this part of the question. The most common error was for candidates to use a distance, for example 19.6 – 5 = 14.6, with the initial speed of 22.4 sinθ.

Q22.

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Q24.

There were many correct answers to part (a) and the candidates were clearly helped by the presence of the printed answer. In most cases the arguments given were sound, but in others this was not the case. For example some candidates simply wrote statements such as 22.4 sin θ – 19.6 and never included an equals sign or a zero. This was regarded as an important omission.
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The approaches to part (d) were many and varied. There were quite a number of good responses, but often candidates did not really get very far with this part of the question.

The most common error was to see candidates use a distance for example 19.6 – 5 = 14.6, with the initial speed of 22.4 sinθ.
In the final part, there was a tendency for the candidates to find the resultant of two velocity components, as they did not realise that for the minimum speed the vertical component would be zero. In some case the correct final answer was obtained from expressions such as $\sqrt{10.8^2 + 0^2}$.

Q25.

This question was also done very well by the candidates. The main issues were in parts (a) and (b). In part (a), a number of candidates made errors when calculating the area under the graph. Often they would have one small error, which meant that their total was incorrect. In these cases, the candidates' distances were followed through into part (b). Both parts (b) and (c) were done well with very few candidates losing marks. In part (d), Newton's second law was applied in all but a few cases, but the issue was with the units of the mass. Some candidates converted to kg incorrectly, while others simply found the force as if the mass were 200 kg.

Q26.

Part (a), and in particular part (a)(i), caused problems for a number of candidates. The issue was that they did not consider the whole system or included a tension force in some inappropriate way. The candidates who found this difficult did not seem to appreciate the need to consider the whole system first. Some candidates did find the tension first and then considered the forces on the car to find P. Part (a)(ii) was done better than part (a)(i), but there was evidence of confusion as to which component of the system to consider and which forces were acting on it.

Part (b) was done very well and a high proportion of the candidates gained full marks on this part.

There were many good answers to part (c), but some candidates did make reference to the condition of the road surface and often included friction as if the car and caravan were sliding on the road surface.

Q27.

Parts (a) and (b) were generally done well, with the candidates gaining many marks. A few candidates could not state the direction in part (b)(ii) and gave answers such as north-east or south-west.Part (c) was found to be more demanding. Candidates who did not identify the **i** and **j** components of the velocity made very little progress. Some made some initial good progress, but made arithmetic or algebraic errors in their solutions. Some forgot to square the speed and tried to work from equations such as $(4 - 0.4t)^2 + (0.5 + 0.2t)^2 = 5$. Some candidates used a trial and improvement approach and realised that the velocity must be −3**i** + 4**j**, which produced a speed of 5.

Q28.

A large proportion of the candidates produced a quadratic equation, but there were several potential errors caused by setting up the equation in the wrong way. A few used the 1 or the 1.5 but not both, some used one with the incorrect sign and ended up with a 2.5 term instead of a 0.5 term, and some used 0.5 but with the wrong sign. Other methods that were seen included finding the time to move up and the time to move down and then adding them together. Some candidates lost one mark due to a lack of intermediate working and a failure to show how 1.30 was obtained from their quadratic equation. Again, showing an answer to more than three significant figures before quoting the final answer would have helped them.

Part (b) was usually done well with candidates using the printed answer from part (a).

In part (c), some candidates used the required approach to find the speed of the arrow, but many did not realise how to approach this part of the question. A few candidates made sign errors so that they obtained a vertical component of velocity that was too great ($12\sin30° + 9.8 \times 1.3 = 18.74$). A few candidates only considered one component of the velocity. Those who had done well on part (c) were usually able to obtain the marks for part (d) too.

There were a lot of good answer to part (e), and in some cases this was the only place on the question where the candidate gained any marks.

Q29.

This question was done well by many candidates. There were a few minor slips in part (a). Part (b) was a little more challenging. The most common errors were to omit one force, typically the weight leading to an answer of 160 N, or to make a sign error, for example $800g - T = 800 \times 0.2$ leading to a final answer of 7680 N.

Q30.

(a) was usually done correctly. Part (b) saw more errors from candidates, with some failing to subtract the 400 from the 2000 and using 2000 metres in their calculations. A few also simply divided their distance by 18 rather than using a constant acceleration equation. In part (c), many candidates drew a graph with the correct shape, but many did not label the required speeds and times clearly on the axes. In many cases, a scale (for example 4, 8, 12, 16,…) had been put on the axis, rather than just indicating the initial and final speeds. Part (d) was generally done very well, but a few candidates did take the mean of 18 and 32 rather than using the total distance divided by the total time.

Q31.

Parts (a), (b) and (c) of this question were done very well. In part (a), almost all candidates used an equation of motion for each particle and there were very few ‘single equation’ solutions. In part (c), some candidates stated one incorrect assumption, for example one about the peg rather than the string. In part (d), a few candidates took the acceleration to be equal to g rather than the value obtained in the question.

Q32.

In part (a), many candidates produced correct answers, but some stopped when they had obtained the velocity and did not go on to find the speed as requested. Part (b) was generally done well, but a few candidates did not give their final answer to the nearest degree. Part (c) was very much more demanding. A significant number of candidates mixed scalar and vector quantities in an attempt to answer the question. Statements like $500 = \frac{1}{2}(0.5\mathbf{i} + 0.375\mathbf{j})t^2$ were quite common and there were some similar ones that also included an initial velocity. The successful candidates used a variety of approaches, which can be seen in the mark scheme. The key difference between those who were successful and those who were not was the ability to connect the scalar distance of 500 to the position vector in a meaningful way.

Q33.

This question was answered very well by the great majority of the candidates. Part (b)(i) of the question was prone to errors for some candidates who did not take sufficient care in getting a correct simplified quadratic equation in $\tan\theta$ and solving the equation to find the two possible values of θ Some of these candidates wrote 89.8° for a possible value of θ and did not seem to question or check the reasonableness of their answer. For part (b)(ii), almost all candidates chose their smaller angle for the answer, but many could not provide a convincing reason for their choice.

Q34.

There was a range of responses to this question, with candidates generally doing better on parts (a), (b) and (c) than on part (d).

In part (a) the most common error was to either include or omit a negative sign in the early working, so that a negative time would have been produced if this had been followed through accurately.

In part (b) some candidates used an equation for the height rather than the velocity, but still equated it to zero. A few candidates ignored the initial height. Very few candidates realised that the ball would have the same speed when it returned to its initial position.

Q35.

Most candidates answered parts (a), (b) and (c) well, although some did get an answer of –4.5**j** for part (b).

In part (c), the candidates who clearly stated and used a constant acceleration equation usually made a good start and completed the question, although there were a few arithmetic errors.

In part (e) many candidates did not attempt to find the magnitude of the force.

Part (f) was generally found more difficult with only a few good attempts. Several candidates did not attempt this part of the question.